

Pattern Recognition 2024: Homework 12

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Problem 1

编号,色泽,根蒂,敲声,纹理,脐部,触感,密度,含糖率的重要性得分为-0.05494505 0.03846154 0.15384615 -0.15384615 -0.15384615 0.15384615 0.23076923 0.1512605 -0.06463505 ,选择前五个重要的feature为'触感', '脐部', '纹理', '敲声', '根蒂'。基于此特征设计的SVM分类器准确率为100%

Problem 2

(i)

$$\begin{aligned} \mathbf{S}_t &= \sum_{i=1}^N (\mathbf{x}^{(i)} - \mathbf{m})(\mathbf{x}^{(i)} - \mathbf{m})^T \\ &= \sum_{j=1}^C \sum_{\mathbf{x} \in \mathcal{X}_j} (\mathbf{x} - \mathbf{m}_j + \mathbf{m}_j - \mathbf{m})(\mathbf{x} - \mathbf{m}_j + \mathbf{m}_j - \mathbf{m})^T \\ &= \mathbf{S}_w + \sum_{j=1}^C \sum_{\mathbf{x} \in \mathcal{X}_j} (\mathbf{m}_j - \mathbf{m})(\mathbf{x} - \mathbf{m}_j)^T + \sum_{j=1}^C \sum_{\mathbf{x} \in \mathcal{X}_j} (\mathbf{x} - \mathbf{m}_j)(\mathbf{m}_j - \mathbf{m})^T + \mathbf{S}_b \\ &= \mathbf{S}_w + \mathbf{S}_b \end{aligned}$$

(ii)

$$\begin{aligned} \widetilde{\mathbf{S}}_w &= \sum_{j=1}^C \sum_{\mathbf{x} \in \mathcal{X}_j} \mathbf{W}^T (\mathbf{x} - \mathbf{m}_j)(\mathbf{x} - \mathbf{m}_j)^T \mathbf{W} \\ \widetilde{\mathbf{S}}_b &= \sum_{j=1}^C N_j \mathbf{W}^T (\mathbf{x}^{(i)} - \mathbf{m})(\mathbf{x}^{(i)} - \mathbf{m}) \mathbf{W} \end{aligned}$$

(iii)

$$J(\mathbf{W}) = \frac{\prod_{diag} \widetilde{\mathbf{S}_b}}{\prod_{diag} \widetilde{\mathbf{S}_w}} = \frac{\prod_{diag} \mathbf{W}^T \mathbf{S}_b \mathbf{W}}{\prod_{diag} \mathbf{W}^T \mathbf{S}_w \mathbf{W}}$$

(iv)

$$\frac{\partial J(\mathbf{W})}{\partial \mathbf{w}_i} = \frac{2\mathbf{w}_i^T \mathbf{S}_b (\mathbf{w}_i^T \mathbf{S}_w \mathbf{w}_i) - 2\mathbf{w}_i^T \mathbf{S}_w (\mathbf{w}_i^T \mathbf{S}_b \mathbf{w}_i)}{(\mathbf{w}_i^T \mathbf{S}_w \mathbf{w}_i)^2} = 0$$

$$\mathbf{w}_i^T \mathbf{S}_b (\mathbf{w}_i^T \mathbf{S}_w \mathbf{w}_i) - \mathbf{w}_i^T \mathbf{S}_w (\mathbf{w}_i^T \mathbf{S}_b \mathbf{w}_i) = 0$$

Problem 3

(i) 图 1与图 2 分别为3D流型和和2D降维之后的结果

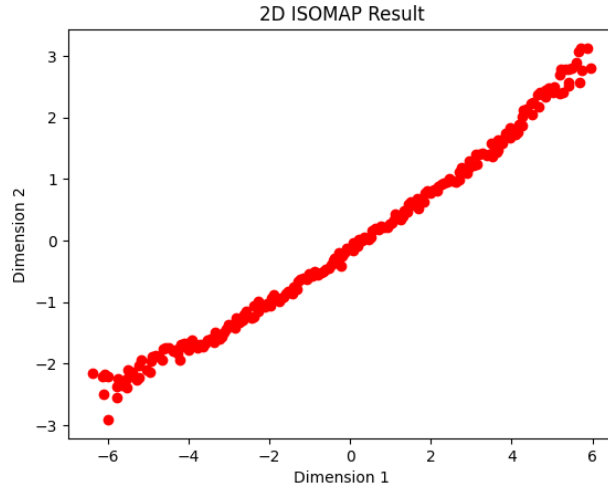


Figure 1: Caption

3D Z-Shaped Manifold

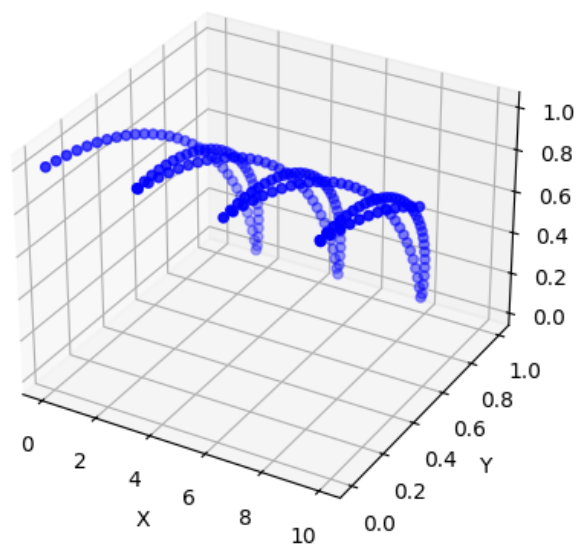


Figure 2: Caption